

# Mini Project: An Explainable and Robust Cyber-Defence Model

AI for Cybersecurity

**Note:** the mini project pulls together everything from all modules, **on a problem you choose yourselves**. You are not asked to invent a new method. You are asked to build a working detector, explain it, test whether it survives with imperfect data, and report all of that honestly. A negative result, reported clearly, is worth more than a good number you cannot defend.

## 1. Choosing your problem and dataset

Your group picks one security problem and one public dataset.

Some options:

- **UNSW-NB15**, network intrusion with a different feature set from CICIDS.
- **NSL-KDD**, the classic intrusion benchmark. Small and fast, but dated, say so in your report.
- **CIC-DDoS2019**, several DDoS attack types, strongly imbalanced.
- **CIC-IoT2023 or Bot-IoT**, traffic from IoT devices.
- **SMS spam / smishing**, text data, so your pipeline will differ from the labs.
- **Your own proposal**, any public, labelled cybersecurity dataset

**These are only suggestions.** You are free to use any public, labelled cybersecurity dataset, including the one used by the paper your group presented at the seminar. Taking that paper's dataset and rebuilding part of its pipeline yourselves is a good way to start, and it means you already know the problem

## 2. What you must build

Parts A to E and Part G are required. Part F is required for grade 5. Make sure your notebook runs from top to bottom in under five minutes. Subsample if you need to, and say that you did.

### Part A. The problem and the data

- State the security problem in two or three sentences: what is being detected, who would use the system, and what a false alarm costs them.
- Load and clean the data. Report how many records and features you have, and the class balance.
- Split 60 / 20 / 20 into training, validation and test, stratified.
- **Report the trivial baseline:** the accuracy of always predicting the majority class, and its recall. Every later score is compared against this.

### Part B. Supervised models

- Train at least one glass box (decision tree or logistic regression) and one black box (random forest or gradient boosting).
- Report macro-F1, recall, PR-AUC and FAR on the test set, next to the trivial baseline.
- **Choose all settings on the validation set.** The test set is touched once, at the end.

### Part C. Learning with fewer labels

- Keep only 5 to 10 per cent of the training labels. Train your black box on those alone. This is your lower line.
- Then try either semi-supervised pseudo-labelling or a self-supervised pretext task followed by a classifier trained on the few labels.
- **Say plainly whether the unlabelled data helped..** A clear negative result, properly explained, scores as well as a positive one.

### Part D. Explanation

- Global: SHAP on your black box. Which features drive it, and do they make sense to a security person?
- Local: explain three cases with SHAP and with LIME, a confident correct detection, a confident correct negative, and a mistake.
- **Fidelity:** run a deletion test. Replace the top SHAP features with training medians and compare the drop with that from the same number of random features.

- **Stability:** run LIME several times with different seeds and report how often the top three features stay the same.

### Part E. Robustness

At least one of the following, reported with numbers at several levels:

- **Noise**, add Gaussian noise scaled to each feature's training standard deviation.
- **Missing features**, replace a fraction of each record's values with the training median.
- **Evasion**, a constrained attack. State which features an attacker could realistically change and in which direction, and keep to it.
- Say what broke first, and at what level you would stop trusting the model.

### Part F. A knowledge-based layer

- Take the two most important features from Part D and build a BRB: three referential values each, nine rules.
- Compare it with a tree and a forest that see only those same two features, so the comparison is fair.
- Show the belief output for one case, including the Unknown part.
- Say what the BRB gives you that the forest does not, and what it costs you.

### Part G. Adaptability

Everything above tests your model on the same kind of data it was trained on. But attacks change. A new variant appears that nobody has seen before, and a model that scored well last month quietly starts missing things. This part measures that, on your own single dataset. You do not need any extra data.

**Step 1. Hide one attack from the model.** Pick one attack type in your data, for example one DDoS variant, one malware family, or one category of fraud. Remove every row of that type from your **training** set only. Keep them in the test set. Now train your black box again on this reduced training set. Your model has never seen this attack.

**Step 2. Measure what it costs.** On the test set, report two numbers: overall recall, and recall on the hidden attack type by itself. The second number is usually very low. That is the point: the model did not get worse at what it knew, the world moved and the model stayed still.

**Step 3. Adapt.** Now imagine an analyst has just labelled a small number of examples of the new attack, say 50 to 100 rows. Add only those to your training set, retrain, and measure the same two numbers again. Report the gain: recall on the new attack before adapting, and after.

**Step 4. Say what you would do.** In one short paragraph: was a small number of labels enough? How would you notice this problem in a real system, where nobody tells you a new attack has appeared?

**If your dataset has only attack and normal labels, with no attack types,** use any column that marks a period, a source or a site instead. Train on some values of that column and test on the rest. If there is nothing like that either, sort the rows by time or by index, train on the first 70 per cent and test on the last 30 per cent. Say in your report which of these you did and why.

## 3. The presentation

**The presentation is on 15 October.** There is no second date and no alternative to presenting.

- **Thirteen minutes per group**, then questions. Keep to time.
- **Every member speaks** and explains the part they built. Divide the work so that this is possible within thirteen minutes.
- **Run your code during the presentation or show a demo of it running.** A slide showing results you obtained **earlier is not enough**. If a full run is too slow, run a reduced version live, or play a recording of the real run and say which it is. (you don't have to use full dataset in you code you can choose (4000 to 5000) rows but make sure there are a mix of attack and normal labels.
- Expect questions about any part of the work, not only your own part. ( you already did all labs work, you don't have to worry about the questions)
- **A member who is absent receives F.** The presentation is held once.

#### 4. What to hand in

**Your code:** one notebook that runs from top to bottom, plus a short README naming the dataset, where to get it, the libraries, and how to run. **A report of 6 to 8 pages** with these headings:

| Heading             | What to write                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Problem and data | The security problem, who uses the system, what a false alarm costs. The dataset, its size, its class balance, and the baseline. And the results before and after validation test                                                                            |
| 2. Method           | Your models, your splits, how settings were chosen, and your approach in Part C.                                                                                                                                                                             |
| 3. Results          | One table with all models and all four metrics, against the trivial baseline. Say whether the unlabelled data helped.                                                                                                                                        |
| 4. Explanation      | The global SHAP picture, your three explained cases, and the fidelity and stability results. Do SHAP and LIME agree?                                                                                                                                         |
| 5. Robustness       | What you tested, at what levels, and what broke first.                                                                                                                                                                                                       |
| 6. Adaptability     | Which attack type you hid, recall on it before and after adapting, and how many labels you needed. How would you notice this in a real system?                                                                                                               |
| 7. Discussion       | Go through Parts A to G in order. For each one, quote your own main result and say what it means, not just what the number is. Then pull it together: would you deploy this? What would an attacker fake? Which of your numbers do you least trust, and why? |
| 8. Who did what     | One or two lines per member.                                                                                                                                                                                                                                 |

#### 5. Deadlines

- Notebook, README and report uploaded; 15 Oct , 8:00 AM
- **Presentation ,15 October** from 14:45 to 18:00

#### Grading

The mini project is graded 5, 4, 3 or F, where 5 is the highest.

| Grade | What the work must show                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | <p>Everything required for grade 4, and:</p> <ul style="list-style-type: none"> <li>Part F is done: a working BRB, compared fairly against models that see the same two features, with the Unknown part shown and discussed.</li> <li>The discussion shows real judgement: which numbers are trustworthy, which are not, what an attacker would fake, and where the method would fail outside this dataset.</li> <li>The explanation work goes past producing plots. The fidelity and stability results are interpreted, and disagreement between SHAP and LIME is investigated rather than ignored.</li> </ul>                                                                                                                                                                                          |
| 4     | <p>Everything required for grade 3, and:</p> <ul style="list-style-type: none"> <li>Part C and Part E are complete, with results reported at several levels rather than a single number.</li> <li>Part G is complete: recall on the hidden attack is reported before and after adapting, and the group says what they would do in a real system.</li> <li>Models are compared properly: the same test set, the same metrics, and the trivial baseline shown alongside.</li> <li>The report offers criticism of its own results, not only a description of them.</li> <li>Whether the unlabelled data helped is answered clearly, including when the answer is no.</li> </ul>                                                                                                                             |
| 3     | <p>All of the following:</p> <ul style="list-style-type: none"> <li>Parts A, B and D are done and correct. The notebook runs from top to bottom and reproduces the numbers in the report.</li> <li>Settings were chosen on the validation set, and the test set was used once.</li> <li>The trivial baseline is reported next to the model scores.</li> <li>Explanations are produced and described at a basic level.</li> <li>Part G is attempted: one attack type is hidden from training and the loss of recall is reported, even if the adaptation step is incomplete.</li> <li>The code was run or demonstrated during the presentation, and every member explained their own part.</li> <li>The report is readable, the figures are captioned, and the who-did-what section is present.</li> </ul> |
| F     | <p>Any of the following:</p> <ul style="list-style-type: none"> <li>Not submitted by the deadline.</li> <li>The member is absent from the presentation on 15 October.</li> <li>The member does not speak during the presentation.</li> <li>The code was neither run nor demonstrated during the presentation.</li> <li>The notebook does not run, or its numbers do not match the report.</li> <li>A required part is missing.</li> <li>The test set was used to choose settings, or the labels of the hidden set were used in training.</li> <li>Work is copied without citation, or AI assistance is used without being declared.</li> </ul>                                                                                                                                                           |

**One attempt only.** There is no resubmission and no second presentation date. The grade awarded after 15 October is final.

#### Submitting your work

- Where.** Upload to CANVAS by the deadline: your code as a zip or a repository link, and the report as a PDF.
- Group work.** Every member contributes to the code, not only to the report. The who-did-what section is read and may affect individual grades.
- Honesty.** You may use libraries, tutorials and public code, just cite them. If you use an AI assistant, say so in the report and describe what it was used for.